

Behzad Ousat

Security Research & ML | Ph.D. Candidate

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📍 Miami, FL - Open to relocate

Security researcher and engineer working on web abuse at scale: bot and phishing detection, Captcha security, and vulnerability analysis across millions of websites and terabytes of traffic. First-author papers at ACM WWW, RAID, and DIMVA, with reviewing service for RAID, DIMVA, and IEEE TIFS. Backed by 4 years operating production infrastructure for a high-traffic ride-hailing platform.

EDUCATION

Ph.D. Computer Science, Florida International University	01/2022 – 12/2026
M.Sc. Computer Engineering, Sharif University of Technology	08/2018 – 08/2021
B.Sc. Computer Engineering, University of Tehran	08/2014 – 08/2018

RESEARCH EXPERIENCE

Florida International University - Research Assistant, Applied ML | Miami, FL 01/2022 – Present

Bot Defenses, Captcha, and Automated Abuse

- Measured Captcha deployment across 1M+ websites, then built an **object detection** pipeline that solved 12 schemes at 80%+ accuracy, exposing a brittle defense (WWW '24).
- Developed **LLM-powered browser agents** that mimic human interaction patterns, reaching 90%+ success on real-world crawling tasks, then used them to re-evaluate commercial bot defenses.
- Trained **LSTM** and **Transformer** models on browser interaction time series to classify automated agents at 95% accuracy, then quantified false-positive rates for production use.
- Attacked behavioral bot detectors with **adversarial ML (FGSM, genetic algorithms)** to establish the practical limits of multi-modal defenses (WWW '24).

Phishing and Supply-Chain Threat Analysis

- Studied development practices and code reuse across **300K+ phishing websites** via a dockerized pipeline that produced 100+ GB of artifacts; joint work with **Microsoft Security** (DIMVA '24).
- Mined **30K reported vulnerabilities** across 10+ package ecosystems with custom crawling pipelines, surfacing CWE patterns, malicious packages, and supply-chain attack trends (RAID '26).
- Benchmarked 10 LLMs for automated vulnerability triage in **CI/CD** pipelines, building an evaluation harness that measured precision, recall, and latency on patched and vulnerable packages (RAID '26).

Intrusion Detection and Data Infrastructure

- Built distributed training and inference workflows over terabytes of **PCAP** data, applying **open set recognition** to flag zero-day network attack classes (JISA '23).
- Operated **multi-modal** collection pipelines that captured 1.6B+ artifacts (DOM, screenshots, event streams) into **NoSQL** storage as ground truth for detection models.

Fraunhofer IDMT - Machine Learning Engineer Intern | Germany 09/2019 – 12/2019

- Built a **face recognition** pipeline (OpenCV, **CNN**) to identify studio scenes across 100+ hours of news video.

INDUSTRY EXPERIENCE

Tapsi - Software and DevOps Engineer | Tehran 12/2017 – 10/2021

Platform Technical Team Lead

- Created **internal platform tooling** (Python, Bash) that let developers run a single microservice locally against shared **Kubernetes** staging environments, cutting local resource needs by 80%.
- Developed **Node.js** and Python message-passing libraries over **gRPC** and **RabbitMQ** that sustained 10K events per second, instrumented with **Prometheus** and **Grafana**.

Senior Infrastructure Engineer

- Migrated 60+ microservices to **Kubernetes**, coordinated with backend teams, and onboarded developers through internal workshops.
- Engineered production data pipelines with **Kafka** and **Apache Spark** to automate collection, processing, and reporting for Python ML microservices.

Infrastructure Engineer

- Maintained 99% uptime across 100+ **Ubuntu** servers running compute clusters, **PostgreSQL**, and **MongoDB** on **Kubernetes**.
- Implemented **CI/CD** pipelines in **Bash** and **Ansible** spanning staging, canary, and production environments, with A/B testing to de-risk releases.

SKILLS

Languages: Python | Node.js | Java | C/C++ | Bash

Security: Bot & Abuse Detection | Phishing | Captcha Security | Vulnerability Research

Methods: Adversarial ML | Open Set Recognition | Time Series | PCAP Analysis

ML: PyTorch | TensorFlow | Transformers | scikit-learn | Pandas | NumPy

Infrastructure: Kubernetes | Docker | Terraform | GitHub Actions | Linux | Ansible

Data & Cloud: PostgreSQL | MongoDB | Kafka | PySpark | Redis | AWS | GCP

SERVICE AND HONORS

Journal Reviewer, IEEE Transactions on Information Forensics & Security (TIFS, IF 8.0) **2024**

Conference Reviewer, RAID, DIMVA, MadWeb **2024, 2025**

Teaching Assistant, Software Security, Operating Systems, Python Programming, FIU **2021–2025**

Dissertation Year Fellowship Award, Florida International University **2026**

PUBLICATIONS

B. Ousat, L. Rampersaud, D. Bailey, N. Turkmen, A. Kharraz, Broken Gates: Re-evaluating Web Bot Defenses in the Age of LLM Agents. Preprint, 2026. arXiv:2607.18659

S. A. Akhavani, **B. Ousat**, A. Kharraz, Vulnerability Evolution and the Promise of Automated Gatekeeping in Open-Source Software. International Symposium on Research in Attacks, Intrusions and Defenses (RAID), 2026.

B. Ousat, M. Shariatnasab, E. Schafir, F. Shirani, A. Kharraz, In-Application Defense Against Evasive Web Scans through Behavioral Analysis. Preprint, 2025. arXiv:2412.07005

B. Ousat, D. Luo, A. Kharraz, Breaking the Bot Barrier: Evaluating Adversarial AI Against Multi-Modal Defenses. ACM Web Conference (WWW), 2024.

B. Ousat, E. Schafir, D. C. Hoang, M. A. Tofighi, S. Arshad, C. Nguyen, S. Uluagac, A. Kharraz, The Matter of Captchas: An Analysis of a Brittle Security Feature on the Modern Web. ACM Web Conference (WWW), 2024.

M. A. Tofighi, **B. Ousat**, J. Zandi, E. Schafir, A. Kharraz, Constructs of Deceit: Exploring Nuances in Modern Social Engineering Attacks. Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA), 2024.

B. Ousat, M. A. Tofighi, A. Kharraz, An End-to-End Analysis of Covid-Themed Scams in the Wild. ACM ASIA Conference on Computer and Communications Security (ASIACCS), 2023.

M. Soltani, **B. Ousat**, M. Jafari Siavoshani, A. H. Jahangir, An Adaptable Deep Learning-based Intrusion Detection System to Zero-day Attacks. Journal of Information Security and Applications, 2023.

L. Rampersaud, **B. Ousat**, S. A. Akhavani, J. Zandi, S. Uluagac, A. Kharraz, Evaluating Security Checks Against Malicious Payloads with Forged Signatures. IEEE IEMCON, 2025.

L. Rampersaud, **B. Ousat**, C. Brown, S. Uluagac, A. Kharraz, On the Effectiveness of End-Users' Data Backup Practices Against Data Corruption. IEEE IEMCON, 2025.